Modifiable Lifestyle Factors

Mechanisms of Action on Physiological Function

SANJAY BHOJRAJ, MD

REGINA DRUZ, MD

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

Disclosures

Regina Druz, MD, was a speaker for Boston Heart Diagnostics.

Sanjay Bhojraj, MD, disclosed no relevant financial relationships with ineligible companies.

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Learning Objectives

At the conclusion of this segment successful participants will be able to:

1. Cite mechanisms of action to explain how physiological dysfunction and clinical imbalances on each node of the Matrix may be caused by modifiable lifestyle factors.

IFM Clinical Skills and Performance Objectives

THIS SEGMENT WILL ENABLE PARTICIPANTS TO DEVELOP & APPLY THESE SKILLS IN THEIR PRACTICE:

1. Prioritize modifiable lifestyle factors as foundational therapeutic interventions when addressing chronic disease.

Clinical Focus

- Modifiable lifestyle factors as antecedents and mediators of physiological dysfunction and clinical imbalances on each node of the matrix.

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Functional Medicine Matrix

The Patient's Story

Antecedents (Inherited & Acquired)

Triggers & Precipitating Events

Mediators/Perpetuators

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Spiritual

Energy

Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships

Name: ______________________ Date: ____________ CC: ____________

How do modifiable lifestyle factors affect the ***Assimilation*** node?

Long Description

- Functional medicine matrix with emphasis on the assimilation node.

Sleep as a Mediator of Dysbiosis

- Suboptimal sleep may indirectly affect maldigestion and malabsorption by disrupting the microbiome.
- Sleep disturbances are also associated with IBS prevalence and other GI conditions, including GERD.
- In an observational study, those with shortened sleep had 32% increased odds of constipation.

See: "References: Mapping Modifiable Lifestyle Factors to the Matrix: Assimilation"

Physical Activity as a Mediator of Dysbiosis

High intensity and strenuous exercise increases stress hormones, which alter nervous system activity, resulting in changes in gut motility, gut transit time (increased usually), and nutrient transport activity.

Sedentary behavior can slow down transit time, increasing the risk of dysbiosis and maldigestion.

Images: Created with BioRender.com by The Institute for Functional Medicine
See: “References: Mapping Modifiable Lifestyle Factors to the Matrix: Assimilation”

Stressors as a Mediator of Dysbiosis

Stress

- Altered motility
- Impaired nutrient absorption
- Sympathetic nervous system activation
- ENS dysfunction
- Decreased stomach acid

Meditation

- Improved cephalic phase of digestion
- Support of downstream digestive processes via activation of the parasympathetic nervous system

See: "References: Mapping Modifiable Lifestyle Factors to the Matrix: Assimilation"

Stressors Alter the Gut Microbiome

- Observational research supports psychosocial stress as a risk factor for psychiatric illness.
- Psychosocial stress may have a similar role in dysbiosis.
- Early life stress alters the enteric microbiome, with effects persisting into adulthood.
- Examples:
  - *Lactobacillus* spp. decreases in response to stress.
  - Patients with major depression had lower abundance of certain genera (*Bifidobacterium, Dialister, Escherichia/Shigella, Faecalibacterium,* and *Ruminococcus*).

See: "References: Mapping Modifiable Lifestyle Factors to the Matrix: Assimilation"

Relationships as a Mediator of Dysbiosis

- Spouses have more similar microbiota and more bacterial taxa in common than siblings (even after accounting for dietary factors).
- Married individuals harbor microbial communities of greater diversity and richness relative to those living alone.
- Couples reporting closer relationships have the greatest diversity.
- Proposed mechanisms:
  - Cohabitation
  - Physical contact
  - Shared lifestyle habits

See: “References: Mapping Modifiable Lifestyle Factors to the Matrix: Assimilation”

Functional Medicine Matrix

THE INSTITUTE FOR FUNCTIONAL MEDICINE®

The Patient's Story

Antecedents (Inherited & Acquired)

Triggers & Precipitating Events

Mediators/Perpetuators

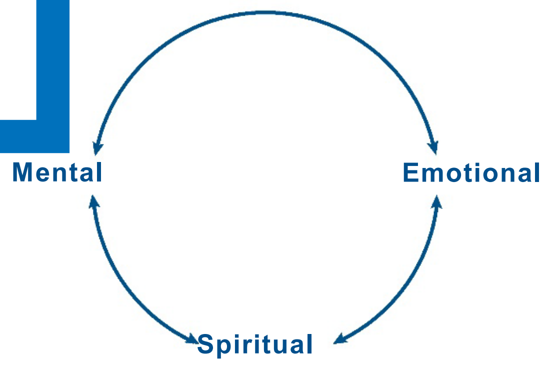
Clinical Imbalances and Physiological Dysfunctions

Assimilation

GERD, Dysbiosis,
Motility Issues,

Defense & Repair

Structural Integrity



Energy

Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships
Poor Sleep	Inactivity		Chronic Stress	Single

Name: ______________________ Date: ____________ CC: ____________

See: "References: Mapping Modifiable Lifestyle Factors to the Matrix: Assimilation"

Long Description

- Functional medicine matrix with GERD, dysbiosis, and motility issues listed under the assimilation node. Poor sleep, inactivity, and chronic stress are shown in the modifiable lifestyle factors.

References

MAPPING MODIFIABLE LIFESTYLE FACTORS TO THE MATRIX: ASSIMILATION

Sleep:

- Karl JP, Whitney CC, Wilson MA, et al. Severe, short-term sleep restriction reduces gut microbiota community richness but does not alter intestinal permeability in healthy young men. *Sci Rep*. 2023;13(1):213. doi:10.1038/s41598-023-27463-0
- Vernia F, Di Ruscio M, Ciccone A, et al. Sleep disorders related to nutrition and digestive diseases: a neglected clinical condition. *Int J Med Sci*. 2021;18(3):593-603. doi:10.7150/ijms.45512
- Zhang G, Wang S, Ma P, et al. Association of habitual sleep duration with abnormal bowel symptoms: a cross-sectional study of the 2005-2010 national health and nutrition examination survey. *J Health Popul Nutr*. 2024;43(1):125. doi:10.1186/s41043-024-00601-8

Physical Activity:

- Costa RJS, Snipe RMJ, Kitic CM, Gibson PR. Systematic review: exercise-induced gastrointestinal syndrome-implications for health and intestinal disease. *Aliment Pharmacol Ther*. 2017;46(3):246-265. doi:10.1111/apt.14157.
- Huang R, Ho SY, Lo WS, Lam TH. Physical activity and constipation in Hong Kong adolescents. *PLoS One*. 2014;9(2):e90193. doi:10.1371/journal.pone.0090193

References

MAPPING MODIFIABLE LIFESTYLE FACTORS TO THE MATRIX: ASSIMILATION

Stressors:

- Cherpak CE. Mindful eating: a review of how the stress-digestion-mindfulness triad may modulate and improve gastrointestinal and digestive function. *Integr Med (Encinitas)*. 2019;18(4):48-53.
- Liu RT. The microbiome as a novel paradigm in studying stress and mental health. *Am Psychol*. 2017;72(7):655-667. doi:10.1037/amp0000058
- Cheung SG, Goldenthal AR, Uhlemann AC, Mann JJ, Miller JM, Sublette ME. Systematic review of gut microbiota and major depression. *Front Psychiatry.* 2019;10:34. doi:10.3389/fpsyt.2019.00034

Relationships:

- Hantsoo L, Jašarević E, Criniti S, et al. Childhood adversity impact on gut microbiota and inflammatory response to stress during pregnancy. *Brain Behav Immun*. 2019;75:240-250. doi:10.1016/j.bbi.2018.11.005
- Dill-McFarland KA, Tang ZZ, Kemis JH, et al. Close social relationships correlate with human gut microbiota composition. *Sci Rep*. 2019;9(1):703. doi:10.1038/s41598-018-37298-9
- Kiecolt-Glaser JK, Wilson SJ, Bailey ML, et al. Marital distress, depression, and a leaky gut: translocation of bacterial endotoxin as a pathway to inflammation. *Psychoneuroendocrinology*. 2018;98:52-60. doi:10.1016/j.psyneuen.2018.08.007

Functional Medicine Matrix

The Patient's Story

Antecedents (Inherited & Acquired)

Triggers & Precipitating Events

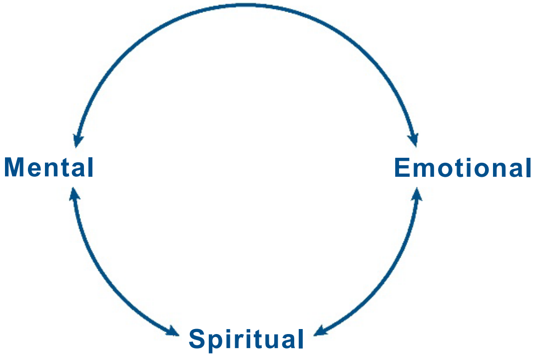
Mediators/Perpetuators

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity



Energy

Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships

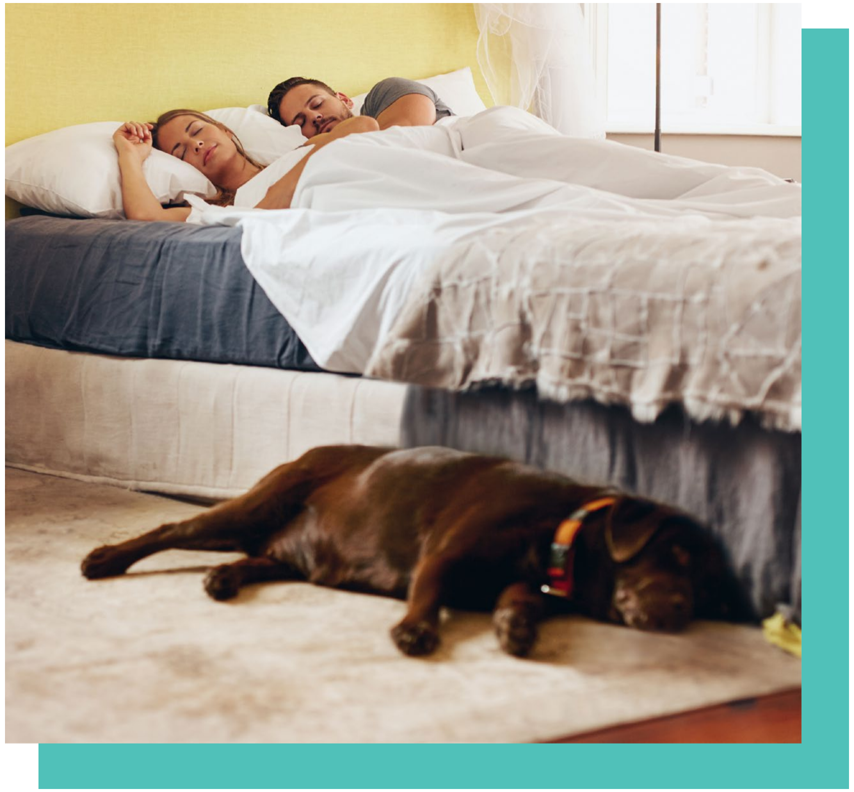
Name: ______________________ Date: __________ CC: __________

How do modifiable lifestyle factors affect the ***Defense & Repair*** node?

Long Description

- Functional medicine matrix with emphasis on the defense and repair node.

Sleep Disturbances Impacting Defense and Repair



- Acute sleep deprivation stimulates inflammatory gene expression and proinflammatory cytokines.
- Evidence suggests that prolonged sleep disturbance leads to increased inflammatory markers, in particular CRP and IL-6.
  - Healthy volunteers randomly assigned to limit sleep to four hours per night for 12 days showed significantly elevated IL-6 compared to controls who slept eight hours per night.

See: "References: Mapping Modifiable Lifestyle Factors to the Matrix: Defense and Repair"

Physical Activity Modulating Defense and Repair

Resistance training increased lean mass and decreased fat percentages while decreasing IL-6 and TNF-a.

Strength training decreased inflammatory cytokines.

Moderate-intensity aerobic exercise and HIIT lowered IL-6 (and possibly hs-CRP) in previously sedentary older men.

See: "References: Mapping Modifiable Lifestyle Factors to the Matrix: Defense and Repair"

Sugar Consumption
Modulation of Pro-inflammatory Mediators

- High intake of dietary sugars leads to metabolic disorders and **inflammatory cytokine production** from adipose tissue and liver.
- High sugar intake may promote **Th17 differentiation** via mitochondrial ROS production.
- Excessive dietary sugar intake reduces the production of short-chain fatty acids in the gut, leading to **increased gut permeability.**
- High dietary sugars lead to **increased TLR4 activity** in macrophages.

See: "References: Mapping Modifiable Lifestyle Factors to the Matrix: Defense and Repair"

Loneliness Modulating Defense and Repair

Loneliness is correlated with ↑ inflammatory cytokines

There was a significant relationship between social disconnection and a heightened TNF-a response in those exposed to an endotoxin group compared to a placebo.

Those sensitive to social disconnection displayed higher IL-6 responses to the endotoxin.
Group with endotoxin exhibited an increase in sensitivity to social disconnection ⟶ to increased IL-6 response.

Loneliness has also been shown to lead to decreased antiviral immunity, increased antibody titers to certain viruses (EBV, CMV, HPV), and decreased immunoglobulin levels (IgA, IgG, IgM).

See: “References: Mapping Modifiable Lifestyle Factors to the Matrix: Defense and Repair”

Functional Medicine Matrix

THE INSTITUTE FOR FUNCTIONAL MEDICINE®

The Patient's Story

Antecedents (Inherited & Acquired)

Triggers & Precipitating Events

Mediators/Perpetuators

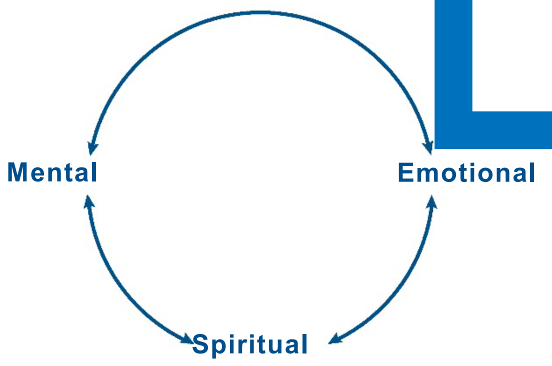
Clinical Imbalances and Physiological

Assimilation

Defense & Repair
Inflammatory dx (high CRP)

Structural Integrity

Energy



Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships
Poor sleep		High-sugar diet		Loneliness

Name: ______________________ Date: ______________ CC: ______________

See: "References: Mapping Modifiable Lifestyle Factors to the Matrix: Defense and Repair"

Long Description

- Functional medicine matrix with inflammatory diagnoses and high C R P listed under the defense and repair node. Poor sleep, high sugar diet, and loneliness are shown in the modifiable lifestyle factors.

References

MAPPING MODIFIABLE LIFESTYLE FACTORS TO THE MATRIX: DEFENSE AND REPAIR

Sleep:

- Irwin MR, Opp MR. Sleep health: reciprocal regulation of sleep and innate immunity. *Neuropsychopharmacology*. 2017;42(1):129-155. doi:10.1038/npp.2016.148
- Irwin MR, Olmstead R, Carroll JE. Sleep disturbance, sleep duration, and inflammation: a systematic review and meta-analysis of cohort studies and experimental sleep deprivation. *Biol Psychiatry.* 2016;80(1):40-52. doi:10.1016/j.biopsych.2015.05.014
- Haack M, Sanchez E, Mullington JM. Elevated inflammatory markers in response to prolonged sleep restriction are associated with increased pain experience in healthy volunteers. *Sleep*. 2007;30(9):1145-1152. doi:10.1093/sleep/30.9.1145

Physical Activity:

- Koh Y, Park KS. Responses of inflammatory cytokines following moderate intensity walking exercise in overweight or obese individuals. *J Exerc Rehabil.* 2017;13(4):472-476. doi:10.12965/jer.1735066.533
- Shultz SP, Dahiya R, Leong GM, Rowlands DS, Hills AP, Byrne NM. Muscular strength, aerobic capacity, and adipocytokines in obese youth after resistance training: a pilot study. *Australas Med J*. 2015;8(4):113-120. doi:10.4066/AMJ.2015.2293
- Hayes LD, Herbert P, Sculthorpe NF, Grace FM. Short-term and lifelong exercise training lowers inflammatory mediators in older men. *Front Physiol*. 2021;12:702248. doi:10.3389/fphys.2021.702248

Nutrition:

- Ma X, Nan F, Liang H, et al. Excessive intake of sugar: an accomplice of inflammation. *Front Immunol*. 2022;13:988481. doi:10.3389/fimmu.2022.988481

References

MAPPING MODIFIABLE LIFESTYLE FACTORS TO THE MATRIX: DEFENSE AND REPAIR

Relationships:

- Pourriyahi H, Yazdanpanah N, Saghazadeh A, Rezaei N. Loneliness: an immunometabolic syndrome. *Int J Environ Res Public Health*. 2021;18(22):12162. doi:10.3390/ijerph182212162
- Walker E, Ploubidis G, Fancourt D. Social engagement and loneliness are differentially associated with neuro-immune markers in older age: time-varying associations from the English Longitudinal Study of Ageing. *Brain Behav Immun*. 2019;82:224-229. doi:10.1016/j.bbi.2019.08.189
- Wilson SJ, Andridge R, Peng J, Bailey BE, Malarkey WB, Kiecolt-Glaser JK. Thoughts after marital conflict and punch biopsy wounds: age-graded pathways to healing. *Psychoneuroendocrinology*. 2017;85:6-13. doi:10.1016/j.psyneuen.2017.07.489

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Functional Medicine Matrix

The Patient's Story

Antecedents (Inherited & Acquired)

Triggers & Precipitating Events

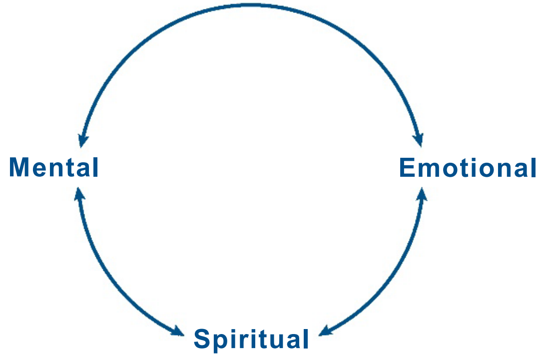
Mediators/Perpetuators

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity



Energy

Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships

Name: ______________________ Date: __________ CC: __________

How do modifiable lifestyle factors affect the ***Communication*** node?

Long Description

- Functional medicine matrix highlighting the communication node.

Sleep Effects on Estrogen

Among a sample of 259 regularly menstruating women:

- Mean estradiol concentrations increased by 3.9% for every hour increase in daily sleep (measured by sleep diaries).
- Mean luteal phase progesterone concentrations increased by 9.4% for every hour increase in daily sleep (measured by sleep diaries).
- Women reporting shift/night work had lower testosterone levels compared to employed women who did not work night or rotating shifts.

Hormonal changes associated with inadequate sleep have the potential to significantly impact gynecological health.

See: "References: Mapping Modifiable Lifestyle Factors to the Matrix: Communication"

Sleep Effects on Testosterone

In an RCT of men with obstructive sleep apnea (OSA) vs healthy volunteers, researchers found:

- An association between OSA and lower levels of total and free testosterone.
- An association between OSA and lower levels of healthy sperm cells and sperm concentration.
- Lower levels of total testosterone after total sleep deprivation.

Partial sleep deprivation did not lead to changes in total testosterone levels.

See: "References: Mapping Modifiable Lifestyle Factors to the Matrix: Communication"

Physical Activity at Work to Reduce Cortisol

N=20 healthy participants, mean age 37.9 years, completed four-week experimental program where they completed regular workdays in a lab setting.

- The participants were randomized to one of three groups:
  - Standing desk
  - Treadmill desk
  - Sitting only
- The participants in the standing desk and treadmill desk groups had significantly lower afternoon salivary cortisol levels compared to their own morning baseline.
- The sitting-only group manifested no significant cortisol changes.

See: "References: Mapping Modifiable Lifestyle Factors to the Matrix: Communication"

Stressful Triggers and TSH

- Among critical care workers and matched controls, those exposed to work-related stress had *significantly higher levels of TSH, LH, and IL-6 and a significantly lower level of FT4* when compared to the control group.

- Long term HPA-axis activation was observed in earthquake-affected populations.
- Increased incidence of biochemical hypercortisolism was identified post earthquake.
- Stressful events triggered HPT-axis activation and higher rates of altered TSH.

See: "References: Mapping Modifiable Lifestyle Factors to the Matrix: Communication"

Stress & Relationships: Mediators of Female Hormone Dysregulation

Adverse Childhood Events (ACEs)

PMS and PMDD

- Women with four or more ACEs are 2.46 times more likely to experience PMDs compared to women with no ACEs.

Early Menarche

- Lower age at menarche was associated with higher odds of experiencing fear, distress, and externalizing disorders as reported by ACE scores.

Single and Recurrent Miscarriage

- Women with three or more ACEs are two times as likely to have a single miscarriage and over three times as likely to experience recurrent miscarriages compared to women with no miscarriage.

Reduced Fecundability and Fertility

- Among women ages 18-45, a study concluded that each ACE increased the risk of fertility difficulties by 9% and reduced fecundability by 32%.

See: “References: Stress & Relationships Altering Women’s Hormones”

References

STRESS & RELATIONSHIPS ALTERING WOMEN'S HORMONES

PMS/PMDD:

- Yang Q, Þórðardóttir EB, Hauksdóttir A, et al. Association between adverse childhood experiences and premenstrual disorders: a cross-sectional analysis of 11,973 women. *BMC Med*. 2022;20(1):60. doi:10.1186/s12916-022-02275-7

Early Menarche:

- Colich NL, Platt JM, Keyes KM, Sumner JA, Allen NB, McLaughlin KA. Earlier age at menarche as a transdiagnostic mechanism linking childhood trauma with multiple forms of psychopathology in adolescent girls. *Psychol Med*. 2020;50(7):1090-1098. doi:10.1017/S0033291719000953

Miscarriage:

- Demakakos P, Linara-Demakakou E, Mishra GD. Adverse childhood experiences are associated with increased risk of miscarriage in a national population-based cohort study in England. *Hum Reprod*. 2020;35(6):1451-1460. doi:10.1093/humrep/deaa113

Fecundability and Fertility:

- Jacobs MB, Boynton-Jarrett RD, Harville EW. Adverse childhood event experiences, fertility difficulties and menstrual cycle characteristics. *J Psychosom Obstet Gynaecol*. 2015;36(2):46-57. doi:10.3109/0167482X.2015.1026892

THE INSTITUTE FOR FUNCTIONAL MEDICINE®

Functional Medicine Matrix

The Patient's Story

Antecedents (Inherited & Acquired)

Triggers & Precipitating Events

Mediators/Perpetuators

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Spiritual

Energy

Communication

Irregular cycles
Hypothyroidism
Low testosterone

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships
Poor sleep			Occupational stressors	High ACE score

Name: ______________________ Date: ______________ CC: ______________

See: "References: Mapping Modifiable Lifestyle Factors to the Matrix: Communication"

Long Description

- Functional medicine matrix with irregular cycles, hypothyroidism, and low testosterone listed under the communication node. Poor sleep, occupational stressors, and high A C E score are shown in the modifiable lifestyle factors.

References

MAPPING MODIFIABLE LIFESTYLE FACTORS TO THE MATRIX: COMMUNICATION

Sleep:

- Alvarenga TA, Fernandes GL, Bittencourt LR, Tufik S, Andersen ML. The effects of sleep deprivation and obstructive sleep apnea syndrome on male reproductive function: a multi-arm randomised trial. *J Sleep Res*. 2023;32(1):e13664. doi:10.1111/jsr.13664
- Michels KA, Mendola P, Schliep KC, et al. The influences of sleep duration, chronotype, and nightwork on the ovarian cycle. *Chronobiol Int*. 2020;37(2):260-271. doi:10.1080/07420528.2019.1694938

Movement:

- Gilson ND, Hall C, Renton A, Ng N, von Hippel W. Do sitting, standing, or treadmill desks impact psychobiological indicators of work productivity? *J Phys Act Health.* 2017;14(10):793-796. doi:10.1123/jpah.2016-0712

Stress:

- Eldin AS, Sabry D, Abdelgwad M, Ramadan MA. Some health effects of work-related stress among nurses working in critical care units. *Toxicol Ind Health.* 2021;37(3):142-151. doi:10.1177/0748233720977413
- Spaggiari G, Setti M, Tagliavini S, et al. The hypothalamic-pituitary-adrenal and -thyroid axes activation lasting one year after an earthquake swarm: results from a big data analysis. *J Endocrinol Invest*. 2021;44(7):1501-1513. doi:10.1007/s40618-020-01457-5

Relationships:

- Hillcoat A, Prakash J, Martin L, et al. Trauma and female reproductive health across the lifecourse: motivating a research agenda for the future of women's health. *Hum Reprod*. 2023;38(8):1429-1444. doi:10.1093/humrep/dead087

Functional Medicine Matrix

The Patient's Story

Antecedents (Inherited & Acquired)

Triggers & Precipitating Events

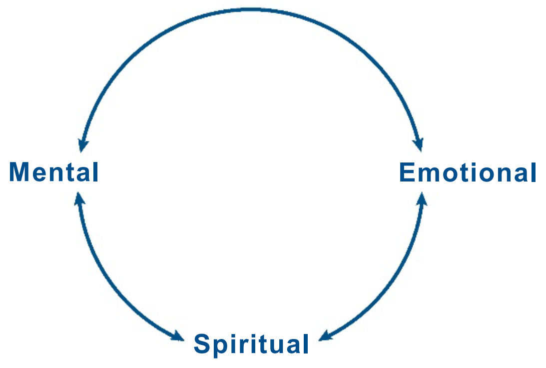
Mediators/Perpetuators

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity



Energy

Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships

Name: ____________________ Date: __________ CC: __________

How do modifiable lifestyle factors affect the ***Transport*** node?

Long Description

- Functional medicine matrix highlighting the transport node.

Inadequate Sleep
Mediator of Transport Dysfunctions

Blood Pressure

- In a systematic review, it was concluded that there was a significant negative association between sleep duration and blood pressure.

Sleep Apnea

- Sleep apnea was significantly associated with sleepiness, inflammation, and insulin resistance; males were significantly sleepier and showed significantly higher hs-CRP, IL-6, leptin, and insulin resistance than controls.

Lipids

- In a meta-analysis, with patients pooled for analysis, patients with obstructive sleep apnea appeared to have increased dyslipidemia (high total cholesterol, LDL, TG, and low HDL).

See: “References: Inadequate Sleep Decreasing Transport”

References

INADEQUATE SLEEP DECREASING TRANSPORT

Blood Pressure:

- Fobian AD, Elliott L, Louie T. A systematic review of sleep, hypertension, and cardiovascular risk in children and adolescents. *Curr Hypertens Rep*. 2018;20(5):42. doi:10.1007/s11906-018-0841-7
- St-Onge MP, Campbell A, Aggarwal B, Taylor JL, Spruill TM, RoyChoudhury A. Mild sleep restriction increases 24-hour ambulatory blood pressure in premenopausal women with no indication of mediation by psychological effects. *Am Heart J*. 2020;223:12-22. doi:10.1016/j.ahj.2020.02.006

Sleep Apnea:

- Kritikou I, Basta M, Vgontzas AN, et al. Sleep apnoea, sleepiness, inflammation and insulin resistance in middle-aged males and females. *Eur Respir J*. 2014;43(1):145-155. doi:10.1183/09031936.00126712

Lipids:

- Nadeem R, Singh M, Nida M, et al. Effect of obstructive sleep apnea hypopnea syndrome on lipid profile: a meta-regression analysis. *J Clin Sleep Med.* 2014;10(5):475-489. doi:10.5664/jcsm.3690

Physical Activity Improves Cardiometabolic Health
Meta Analyses

Blood Pressure

- Aerobic exercise significantly reduced systolic blood pressure (-8.29 mmHg), diastolic blood pressure (-5.19 mmHg), and heart rate (-4.22 beats per minute).
- Interventions including yoga asanas were found to decrease resting SBP by 5.34 mmHg and decrease resting DBP by 3.66 mmHg compared to active controls.

Lipids

- Compared to controls, exercise groups had lower levels of triglycerides and higher levels of high-density lipoprotein cholesterol and apolipoprotein A1.

Metabolic Syndrome/T2DM

- Both supervised aerobic and supervised resistance exercises showed a significant reduction in HbA1c compared to no exercise (0.30% lower, 0.30% lower, respectively).
- Combined (aerobic and resistance) exercise produced even better HbA1c control compared to aerobic alone or resistance alone (HbA1c in combined training groups 0.17% lower, 0.23% lower, respectively). Supervised aerobic also presented more significant improvement than no exercise in fasting plasma glucose (9.38 mg/dl lower).

See: “References: Several Meta-Analyses Conclude Exercise Improves Cardiometabolic Health”

References

SEVERAL META-ANALYSES CONCLUDE EXERCISE IMPROVES CARDIOMETABOLIC HEALTH

Blood Pressure:

- Lee SH, Chae YR. Characteristics of aerobic exercise as determinants of blood pressure control in hypertensive patients: a systematic review and meta-analysis. *J Korean Acad Nurs*. 2020;50(6):740-756. doi:10.4040/jkan.20169
- Pascoe MC, Thompson DR, Ski CF. Yoga, mindfulness-based stress reduction and stress-related physiological measures: a meta-analysis. *Psychoneuroendocrinology*. 2017;86:152-168. doi:10.1016/j.psyneuen.2017.08.008

Lipids:

- Lin X, Zhang X, Guo J, et al. Effects of exercise training on cardiorespiratory fitness and biomarkers of cardiometabolic health: a systematic review and meta-analysis of randomized controlled trials. *J Am Heart Assoc.* 2015;4(7):e002014. doi:10.1161/JAHA.115.002014

Metabolic Syndrome/T2DM:

- Pan B, Ge L, Xun YQ, et al. Exercise training modalities in patients with type 2 diabetes mellitus: a systematic review and network meta-analysis. *Int J Behav Nutr Phys Act.* 2018;15(1):72. doi:10.1186/s12966-018-0703-3

Association of Perceived Stress and Hypertension

Cohort Study of 1,829 Black Participants Without HTN at Baseline

- During follow up (median seven years), hypertension incidence was 48.5%.
- In this community cohort, higher perceived stress was associated with higher risk of developing hypertension:
  - In individuals with <u>low</u> perceived stress between exam intervals, hypertension incidence was 30.6%.
  - In individuals with <u>moderate</u> perceived stress between exam intervals, hypertension incidence was 34.6% (RR = 1.19 compared to low-stress group).
  - In individuals with <u>high</u> perceived stress between exam intervals, hypertension incidence was 38.2% (RR = 1.37 compared to low-stress group).

See: “References: Mapping Modifiable Lifestyle Factors to the Matrix: Transport”

Loneliness Is Predictive of Higher Blood Pressure

- Greater levels of loneliness observed to be predictive of exaggerated blood pressure and inflammation responses to acute stress.
- Systematic review: loneliness associated with higher blood pressure at baseline.
- Lonely individuals also have lower cardiac output and heart rate compared to non-lonely people, indicating that loneliness may cause a blunted response to stress compared to non-lonely people.
- Lonely individuals may perceive certain stressors as more stressful than others.

See: “References: Mapping Modifiable Lifestyle Factors to the Matrix: Transport”

THE INSTITUTE FOR FUNCTIONAL MEDICINE®

Functional Medicine Matrix

The Patient's Story

Antecedents (Inherited & Acquired)

Triggers & Precipitating Events

Mediators/Perpetuators

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Energy

Spiritual

Communication

Mitigation & Clearance

Hypertension

Transport

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships
Poor sleep	Lack of exercise		Multiple stressors	Loneliness

Name: ______________________ Date: ____________ CC: ____________

See: "References: Mapping Modifiable Lifestyle Factors to the Matrix: Transport"

Long Description

- Functional medicine matrix with hypertension in the transport node. Poor sleep, lack of exercise, multiple stressors, and loneliness are shown in the modifiable lifestyle factors.

References

MAPPING MODIFIABLE LIFESTYLE FACTORS TO THE MATRIX: TRANSPORT

Sleep:

- Fobian AD, Elliott L, Louie T. A systematic review of sleep, hypertension, and cardiovascular risk in children and adolescents. *Curr Hypertens Rep*. 2018;20(5):42. doi:10.1007/s11906-018-0841-7

Movement:

- Lee SH, Chae YR. Characteristics of aerobic exercise as determinants of blood pressure control in hypertensive patients: a systematic review and meta-analysis. *J Korean Acad Nurs*. 2020;50(6):740-756. doi:10.4040/jkan.20169

Stress:

- Spruill TM, Butler MJ, Thomas SJ, et al. Association between high perceived stress over time and incident hypertension in black adults: findings from the Jackson Heart Study. *J Am Heart Assoc.* 2019;8(21):e012139. doi:10.1161/JAHA.119.012139

Relationships:

- Brown EG, Gallagher S, Creaven AM. Loneliness and acute stress reactivity: a systematic review of psychophysiological studies. *Psychophysiology.* 2018;55(5):e13031. doi:10.1111/psyp.13031

Modifiable Lifestyle Factors: Part 2

Mechanisms of Action on Physiological Function

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE

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Functional Medicine Matrix

The Patient's Story

Antecedents (Inherited & Acquired)

Triggers & Precipitating Events

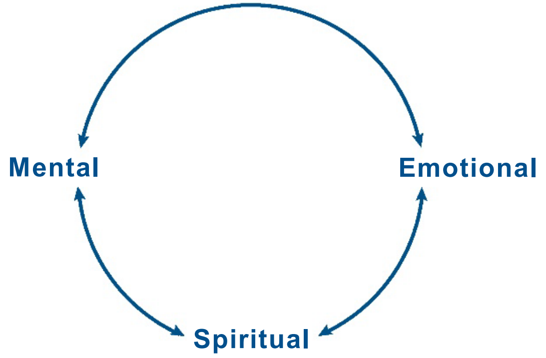
Mediators/Perpetuators

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity



Energy

Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships

Name: ____________________ Date: __________ CC: __________

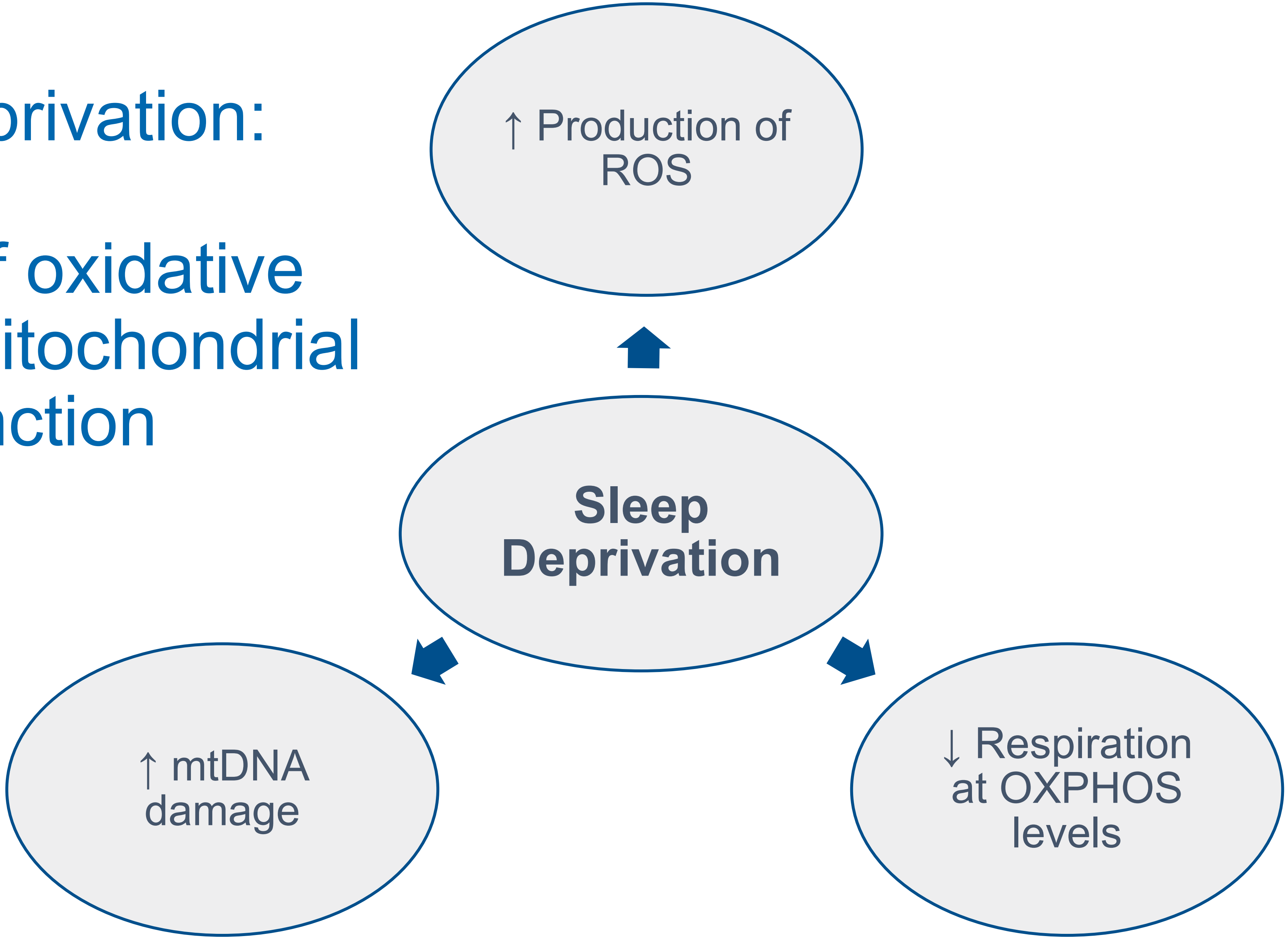
How do modifiable lifestyle factors affect the ***Energy*** node?

Long Description

- Functional medicine matrix highlighting the energy node.

Sleep Deprivation:

Mediator of oxidative stress and mitochondrial dysfunction



See: “References: Mapping Modifiable Lifestyle Factors to the Matrix: Energy”

Long Description

- Sleep deprivation leads to increased production of R O S, decreased respiration at OX PHOS levels, and increased mitochondrial D N A damage.

High-Intensity Exercise Improves Mitochondrial Function

- In a randomized six-week trial comparing high-intensity exercise training or regular lifestyle regime, 20 mildly fit men completed a study.
- High-intensity exercise training increased maximal mitochondrial oxidative phosphorylation (ATP production) capacity and efficiency.

Mitochondrial health is impaired with aging but partially restored with regular exercise. Both moderate and high-intensity aerobic exercise as well as resistance training stimulate biogenesis of mitochondria in skeletal muscle.

See: “References: Mapping Modifiable Lifestyle Factors to the Matrix: Energy”

Effects of Phytonutrient Density on Mitochondrial Health

- Higher consumption of vegetables is related to significantly higher mean telomere length.
- Various pathways of inflammation, insulin, and stress response may be favourably modulated by phytochemicals.
- Antioxidant-poor diets activate the inflammasome, which is a critical convergence point for stress and nutritional dysregulation.

See: "References: Mapping Modifiable Lifestyle Factors to the Matrix: Energy"

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Functional Medicine Matrix

The Patient's Story

Antecedents (Inherited & Acquired)

Triggers & Precipitating Events

Mediators/Perpetuators

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Spiritual

Energy

Fatigue

Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships
Poor sleep	Sedentary behavior	Ultra-processed diet		

Name: ____________________ Date: __________ CC: __________

See: "References: Mapping Modifiable Lifestyle Factors to the Matrix: Energy"

Long Description

- Functional medicine matrix with fatigue in the energy node. Poor sleep, sedentary behavior, and ultra-processed diet are shown in the modifiable lifestyle factors.

References

MAPPING MODIFIABLE LIFESTYLE FACTORS TO THE MATRIX: ENERGY

Sleep:

- Melhuish Beaupre LM, Brown GM, Braganza NA, Kennedy JL, Gonçalves VF. Mitochondria's role in sleep: novel insights from sleep deprivation and restriction studies. *World J Biol Psychiatry*. 2022;23(1):1-13. doi:10.1080/15622975.2021.1907723
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- Rodrigues NR, Macedo GE, Martins IK, et al. Short-term sleep deprivation with exposure to nocturnal light alters mitochondrial bioenergetics in *Drosophila*. *Free Radic Biol Med*. 2018;120:395-406. doi:10.1016/j.freeradbiomed.2018.04.549
- Trivedi MS, Holger D, Bui AT, Craddock TJA, Tartar JL. Short-term sleep deprivation leads to decreased systemic redox metabolites and altered epigenetic status. *PLoS One*. 2017;12(7):e0181978. doi:10.1371/journal.pone.0181978
- Zhang W, Liu D, Yuan M, Zhu LQ. The mechanisms of mitochondrial abnormalities that contribute to sleep disorders and related neurodegenerative diseases. *Ageing Res Rev*. 2024;97:102307. doi:10.1016/j.arr.2024.102307

Physical Activity:

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- Tanaka T, Nishimura A, Nishiyama K, Goto T, Numaga-Tomita T, Nishida M. Mitochondrial dynamics in exercise physiology. *Pflugers Arch*. 2020;472(2):137-153. doi:10.1007/s00424-019-02258-3

References

MAPPING MODIFIABLE LIFESTYLE FACTORS TO THE MATRIX: ENERGY

Nutrition:

- Marcon F, Siniscalchi E, Crebelli R, et al. Diet-related telomere shortening and chromosome stability. *Mutagenesis*. 2012;27(1):49-57. doi:10.1093/mutage/ger056
- Towers AE, Freund GG. Nutritional psychoneuroimmunology: is the inflammasome a critical convergence point for stress and nutritional dysregulation? *Curr Opin Behav Sci*. 2019;28:20-24. doi:10.1016/j.cobeha.2019.01.014
- Vincent HK, Bourguignon CM, Taylor AG. Relationship of the dietary phytochemical index to weight gain, oxidative stress and inflammation in overweight young adults. *J Hum Nutr Diet*. 2010;23(1):20-29. doi:10.1111/j.1365-277X.2009.00987.x

Functional Medicine Matrix

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The Patient's Story

Antecedents (Inherited & Acquired)

Triggers & Precipitating Events

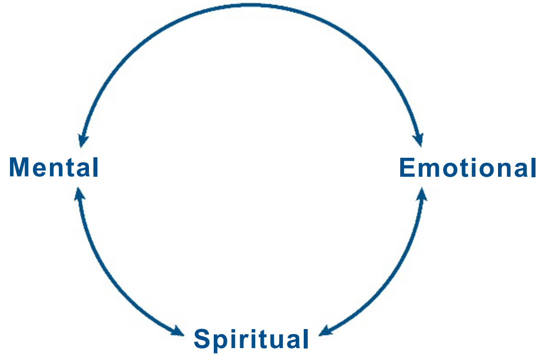
Mediators/Perpetuators

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity



Energy

Communication

$\boxed{\textbf{Mitigation \& Clearance}}$

Transport

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships

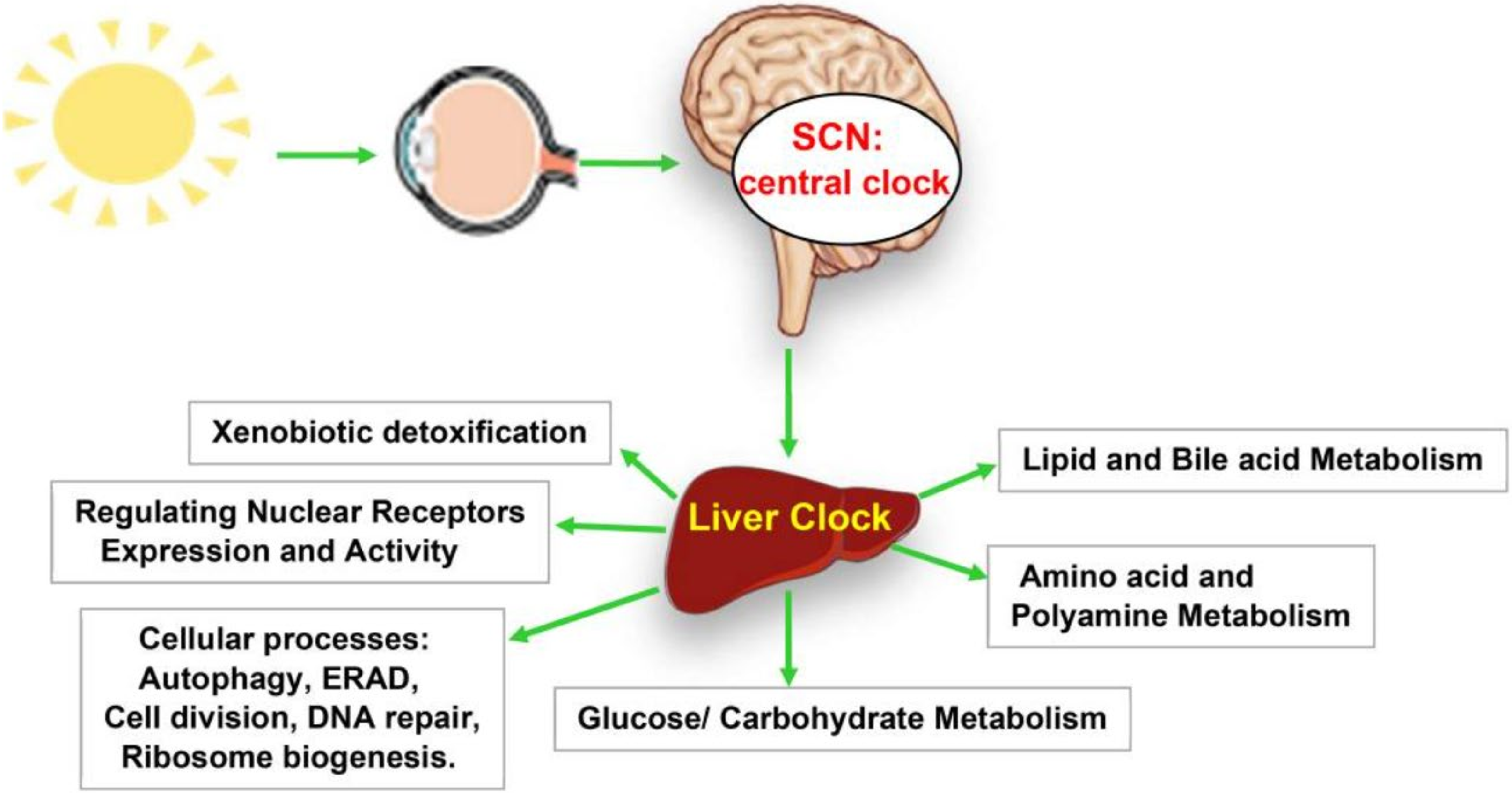
Name: ______________________ Date: ____________ CC: ____________

How do modifiable lifestyle factors affect the ***Mitigation & Clearance*** node?

Long Description

- Functional medicine matrix highlighting the mitigation and clearance node.

Sleep and Circadian Effects on Liver Function



Reprinted from Mukherji A, Bailey SM, Staels B, Baumert TF. The circadian clock and liver function in health and disease. J Hepatol. 2019 Jul;71(1):200-211. doi: 10.1016/j.jhep.2019.03.020. Epub 2019 Mar 28. PMID: 30930223; PMCID: PMC7613420 with permission from Elsevier.

Long Description

- The suprachiasmatic nucleus receives light signaling through the eye, coordinating the circadian rhythms of other organ clocks in the body, such as the liver clock, which regulates various processes including xenobiotic detoxification, cellular processes, and energy metabolism.

Sleep and Circadian Disruption

Effects on Metabolism and Liver Disease

- Contemporary lifestyle, which may include irregular sleep patterns, consistent exposure to artificial light, irregular eating patterns, and rotating night shifts can contribute to disrupted circadian rhythms also known as circadian misalignment.

- The circadian clock is essential for regulating normal hepatic and cellular functions, and its disruption is associated with higher risk for chronic diseases such as obesity, non-alcoholic fatty liver disease (NAFLD), non-alcoholic steatohepatitis (NASH), type 2 diabetes, and metabolic syndrome.

See: "References: Mapping Modifiable Lifestyle Factors to the Matrix: Mitigation & Clearance"

Physical Activity & Elimination

Physical activity can induce mobilization of xenobiotics and toxic metals via sweat.

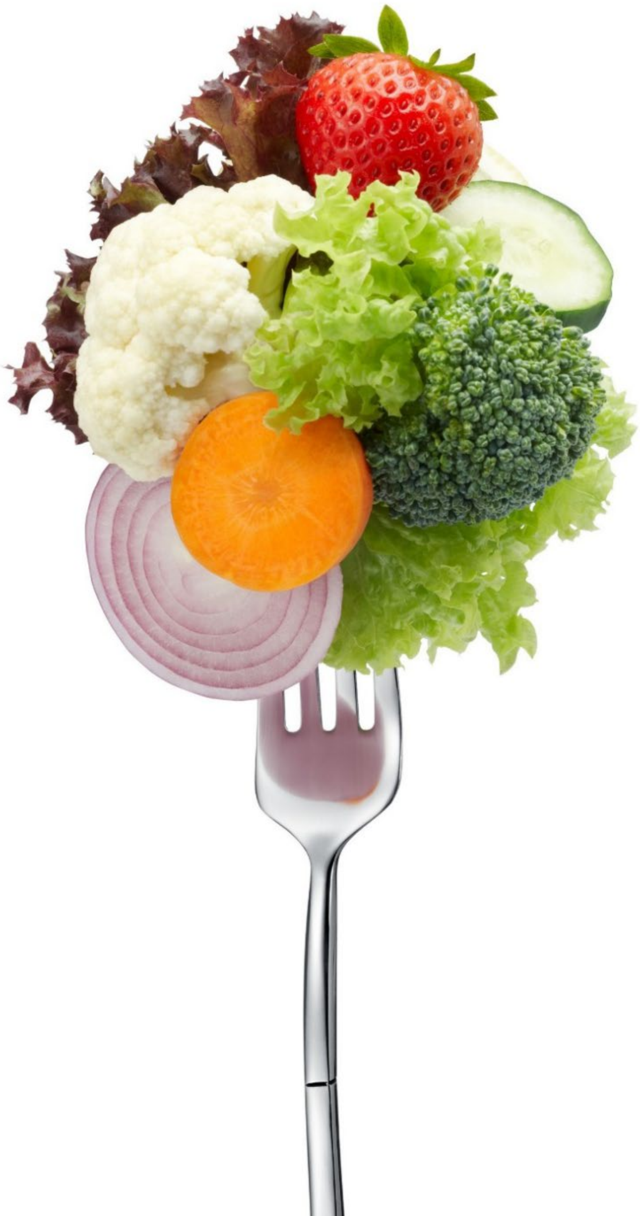
Studies have demonstrated that toxic metals such as **arsenic, cadmium, bismuth, antimony, tin, nickel, lead, & mercury** have been detected in sweat, suggesting that lifestyle factors such as physical exercise can support detoxification.

See: “References: Mapping Modifiable Lifestyle Factors to the Matrix: Mitigation & Clearance”

Nutrition and Detoxification

Research suggests that diets high in antioxidant-rich fruits and vegetables as well as fiber may support detoxification. These include:

- Cruciferous vegetables: watercress, garden cress, and broccoli
- Allium vegetables
- Apiaceous vegetables: carrots, celery
- Grapefruit
- Resveratrol
- Fish oil
- Quercetin
- Daidzein (soy)
- Lycopene



See: “References: Mapping Modifiable Lifestyle Factors to the Matrix: Mitigation & Clearance”

Stress & Anxiety Adversely Affect Liver Function

Stress can significantly affect biotransformation by altering the activity of enzymes, primarily cytochrome P450s (CYPs) in the liver.

- Stress-induced liver injury
- Elevated liver enzymes
- HPA axis activation
- Oxidative stress

See: "References: Mapping Modifiable Lifestyle Factors to the Matrix: Mitigation & Clearance"

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Functional Medicine Matrix

The Patient's Story

Antecedents (Inherited & Acquired)

Triggers & Precipitating Events

Mediators/Perpetuators

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Spiritual

Energy

Communication

Mitigation & Clearance

High liver enzymes
Lives in polluted area

Transport

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships
Shift work (3rd)	Inactive	Ultra-processed diet	High perceived stress	

Name: ____________________ Date: ____________ CC: ____________

See: "References: Mapping Modifiable Lifestyle Factors to the Matrix: Mitigation & Clearance"

Long Description

- Functional medicine matrix with high liver enzymes and polluted environment in the mitigation and clearance node. Shift work, inactivity, ultra-processed diet, and high perceived stress are shown in the modifiable lifestyle factors.

References

MAPPING MODIFIABLE LIFESTYLE FACTORS TO THE MATRIX: MITIGATION AND CLEARANCE

Sleep:

- Mukherji A, Bailey SM, Staels B, Baumert TF. The circadian clock and liver function in health and disease. *J Hepatol.* 2019;71(1):200-211. doi:10.1016/j.jhep.2019.03.020
- Singh A, Anjum B, Naz Q, et al. Night shift-induced circadian disruption: links to initiation of non-alcoholic fatty liver disease/non-alcoholic steatohepatitis and risk of hepatic cancer. *Hepatoma Res*. Published online October 30, 2024. doi:10.20517/2394-5079.2024.88
- Demori I, Grasselli E. The role of the stress response in metabolic dysfunction-associated fatty liver disease: a psychoneuroendocrineimmunology-based perspective. *Nutrients.* 2023;15(3):795. doi:10.3390/nu15030795
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Physical Activity:

- Genuis SJ, Birkholz D, Rodushkin I, Beesoon S. Blood, urine, and sweat (BUS) study: monitoring and elimination of bioaccumulated toxic elements. *Arch Environ Contam Toxicol.* 2011;61(2):344-357. doi:10.1007/s00244-010-9611-5
- Kuan WH, Chen YL, Liu CL. Excretion of Ni, Pb, Cu, As, and Hg in sweat under two sweating conditions. *Int J Environ Res Public Health*. 2022;19(7):4323. doi:10.3390/ijerph19074323
- Sears ME, Kerr KJ, Bray RI. Arsenic, cadmium, lead, and mercury in sweat: a systematic review. *J Environ Public Health.* 2012;2012:184745. doi:10.1155/2012/184745
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References

MAPPING MODIFIABLE LIFESTYLE FACTORS TO THE MATRIX: MITIGATION AND CLEARANCE

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- Kieffer DA, Martin RJ, Adams SH. Impact of dietary fibers on nutrient management and detoxification organs: gut, liver, and kidneys. *Adv Nutr*. 2016;7(6):1111-1121. doi:10.3945/an.116.013219
- Li Z, Wu J, Zhao Y, Song J, Wen Y. Natural products and dietary interventions on liver enzymes: an umbrella review and evidence map. *Front Nutr*. 2024;11:1300860. doi:10.3389/fnut.2024.1300860
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- Sousa RMOF, Cunha AC, Fernandes-Ferreira M. The potential of Apiaceae species as sources of singular phytochemicals and plant-based pesticides. *Phytochemistry*. 2021;187:112714. doi:10.1016/j.phytochem.2021.112714

Stress and Anxiety:

- Adam EK, Quinn ME, Tavernier R, McQuillan MT, Dahlke KA, Gilbert KE. Diurnal cortisol slopes and mental and physical health outcomes: a systematic review and meta-analysis. *Psychoneuroendocrinology.* 2017;83:25-41. doi:10.1016/j.psyneuen.2017.05.018
- Ishtiaq SM, Khan JA, Arshad MI. Psychosocial-stress, liver regeneration and weight gain: a conspicuous pathophysiological triad. *Cell Physiol Biochem.* 2018;46(1):1-8. doi:10.1159/000488378
- Joung JY, Cho JH, Kim YH, Choi SH, Son CG. A literature review for the mechanisms of stress-induced liver injury. *Brain Behav*. 2019;9(3):e01235. doi:10.1002/brb3.1235
- Konstandi M, Johnson EO, Lang MA. Consequences of psychophysiological stress on cytochrome P450-catalyzed drug metabolism. *Neurosci Biobehav Rev*. 2014;45:149-167. doi:10.1016/j.neubiorev.2014.05.011
- Takahashi K, Yamada T, Tsukita S, et al. Chronic mild stress alters circadian expressions of molecular clock genes in the liver. *Am J Physiol Endocrinol Metab.* 2013;304(3):E301-E309. doi:10.1152/ajpendo.00388.2012

THE INSTITUTE FOR FUNCTIONAL MEDICINE®

Functional Medicine Matrix

The Patient's Story

Antecedents (Inherited & Acquired)

Triggers & Precipitating Events

Mediators/Perpetuators

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Spiritual

Energy

Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships

Name: ____________________ Date: __________ CC: __________

How do modifiable lifestyle factors affect the ***Structural Integrity*** node?

Long Description

- Functional medicine matrix highlighting the structural integrity node.

Loss of Sleep Amplifies Pain

Those experiencing sleep deprivation had significantly lower pain thresholds compared to those with adequate sleep.

Depressed Mood or Anger ⟷ Sleep Deprivation

Depressed Mood or Anger ⟷ Reduced Pain Threshold

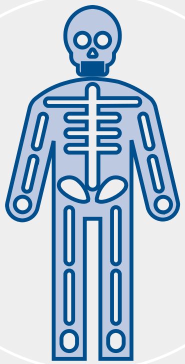
Sleep Deprivation ⟷ Reduced Pain Threshold

See: "References: Mapping Modifiable Lifestyle Factors to the Matrix: Structural Integrity"

Effects of Physical Inactivity on Structural Integrity

Extracellular Matrix (ECM)

- Physical inactivity can lead to ECM degradation, compromising tissue integrity.
- Regular physical activity stimulates ECM production and turnover, maintaining tissue health.



Connective Tissue Health

- A sedentary lifestyle may result in reduced muscle mass and vascularization, affecting connective tissue health.
- Prolonged sitting has been linked to weakened periosteal attachment and bone density, further compromising structural integrity.

Fascia Health

- Inactivity can lead to unhealthy fascia – the connective tissue surrounding muscles and organs – resulting in stiffness and pain.
- Maintaining an active lifestyle is crucial for fascia health

See: “References: Mapping Modifiable Lifestyle Factors to the Matrix: Structural Integrity”

Nutrition and Structural Integrity

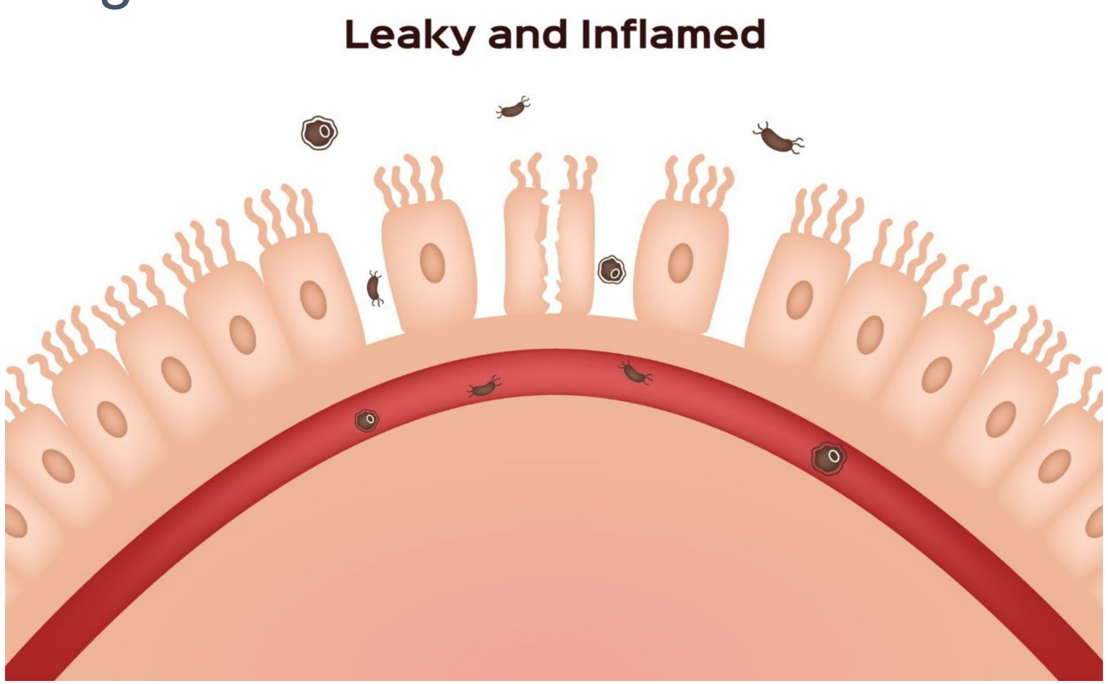
- Diets rich in phytonutrients have been shown to reduce osteoarthritis pain.

- Daily intake of blueberries appears to improve the symptoms of osteoarthritis, including improving walking gait and function.

See: “References: Mapping Modifiable Lifestyle Factors to the Matrix: Structural Integrity”

Hostile Relationships and Intestinal Permeability

- A compromised intestinal barrier can allow bacterial endotoxins (LPS) to enter systemic circulation and is a known marker for intestinal permeability.
- In an RCT, participants involved in hostile marital interactions had higher LPS-binding protein compared to marital partners without discord.



Conclusions:

Distressed marriage and history of depression contribute to a pro-inflammatory state by increasing gut permeability and fueling other inflammation-related disorders.

See: "References: Mapping Modifiable Lifestyle Factors to the Matrix: Structural Integrity"

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Functional Medicine Matrix

The Patient's Story

Antecedents (Inherited & Acquired)

Triggers & Precipitating Events

Mediators/Perpetuators

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity

Mental

Emotional

Spiritual

Energy

Osteoporosis

Increased intestinal permeability

Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships
Inadequate sleep	Inactive			Marital distress

Name: ____________ Date: ____________ CC: ____________

See: "References: Mapping Modifiable Lifestyle Factors to the Matrix: Structural Integrity"

Long Description

- Functional medicine matrix with osteoporosis and increased intestinal permeability in the structural integrity node. Inadequate sleep, inactivity, and marital distress are shown in the modifiable lifestyle factors.

References

MAPPING MODIFIABLE LIFESTYLE FACTORS TO THE MATRIX: STRUCTURAL INTEGRITY

Sleep:

- Krause AJ, Prather AA, Wager TD, Lindquist MA, Walker MP. The pain of sleep loss: a brain characterization in humans. *J Neurosci.* 2019;39(12):2291-2300. doi:10.1523/JNEUROSCI.2408-18.2018

Physical Activity:

- Garg K, Boppart MD. Influence of exercise and aging on extracellular matrix composition in the skeletal muscle stem cell niche. *J Appl Physiol (1985)*. 2016;121(5):1053-1058. doi:10.1152/japplphysiol.00594.2016
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- Schleip R, Gabbiani G, Wilke J, et al. Fascia is able to actively contract and may thereby influence musculoskeletal dynamics: a histochemical and mechanographic investigation. *Front Physiol*. 2019;10:336. doi:10.3389/fphys.2019.00336

Nutrition:

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- Schell J, Scofield RH, Barrett JR, et al. Strawberries improve pain and inflammation in obese adults with radiographic evidence of knee osteoarthritis. *Nutrients*. 2017;9(9):949. doi:10.3390/nu9090949

References

MAPPING MODIFIABLE LIFESTYLE FACTORS TO THE MATRIX: STRUCTURAL INTEGRITY

Relationships:

- Kiecolt-Glaser JK, Wilson SJ, Bailey ML, et al. Marital distress, depression, and a leaky gut: translocation of bacterial endotoxin as a pathway to inflammation. *Psychoneuroendocrinology*. 2018;98:52-60. doi:10.1016/j.psyneuen.2018.08.007

Clinical Focus

- Mechanisms of action of mental, emotional, and spiritual factors as mediators of physiological dysfunction.

Functional Medicine Matrix

THE INSTITUTE FOR FUNCTIONAL MEDICINE®

The Patient's Story

Antecedents (Inherited & Acquired)

Triggers & Precipitating Events

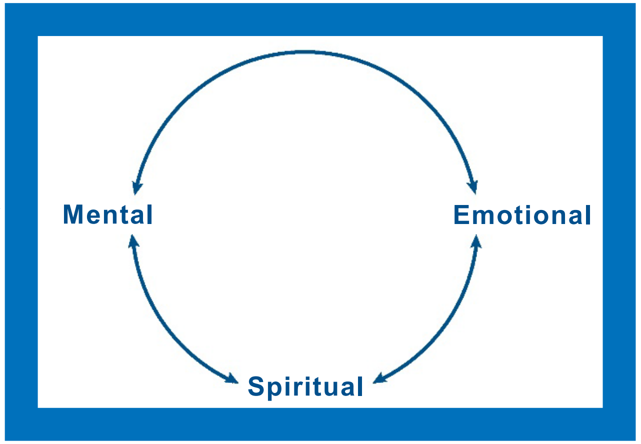
Mediators/Perpetuators

Clinical Imbalances and Physiological Dysfunctions

Assimilation

Defense & Repair

Structural Integrity



Energy

Communication

Mitigation & Clearance

Transport

Modifiable Lifestyle Factors

Sleep	Physical Activity	Nutrition	Stressors	Relationships

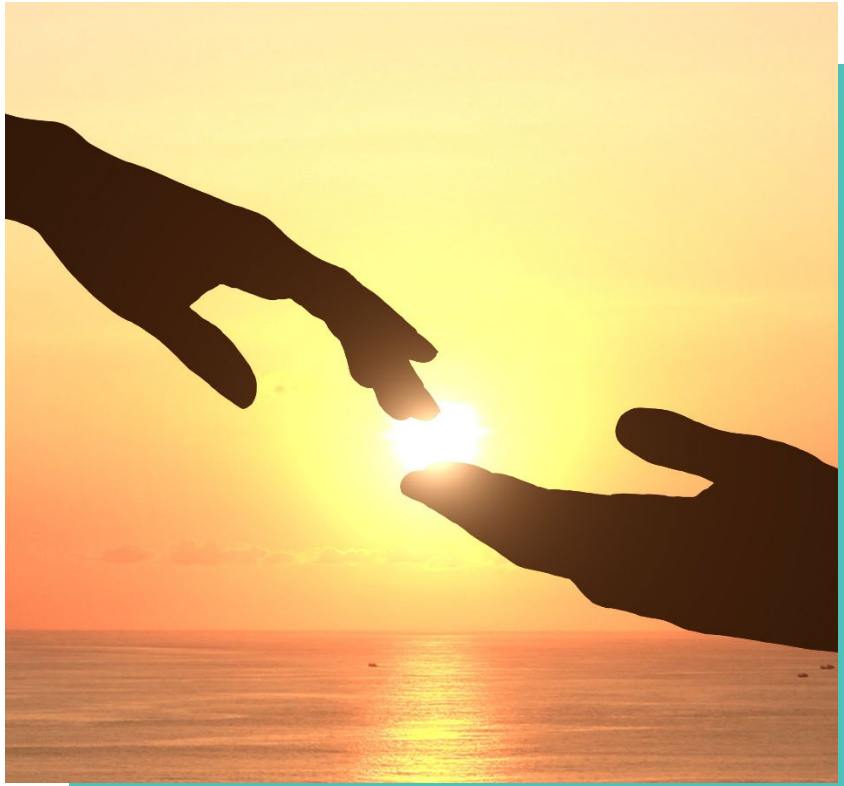
Name: ______________________ Date: __________ CC: __________

Mental, emotional, and spiritual factors.

Long Description

- Functional medicine matrix emphasizing the mental, emotional, spiritual node at the center of the matrix.

Spirituality Practices
Improved Quality of Life: Cancer Patients



Several studies have concluded that incorporating a spiritual practice into the treatment of cancer greatly improves quality of life and coping abilities, both for the patients and their families.

- Breast Cancer
- Gastrointestinal Cancer
- Brain Cancer
- Lung Cancer

Systematic review of 53 studies:

- There is a significant role of spirituality and how it plays a role in coping with difficult diagnoses.

Nagy DS, Isaic A, Motofelea AC, Popovici DI, Diaconescu RG, Negru SM. The role of spirituality and religion in improving quality of life and coping mechanisms in cancer patients. *Healthcare (Basel)*. 2024;12(23):2349. doi:10.3390/healthcare1223234

Center of Matrix and Mitochondria

Numerous studies have concluded that mitochondrial dysfunctions play a role in psychiatric disease such as bipolar disorder.

Dysfunctions in the mitochondria release reactive oxygen species.

Reports conclude that using cognitive behavioral therapies can help mitigate mitochondrial dysfunction and reduce reactive oxygen species.

Targeting mitochondrial dysfunction by reducing reactive oxygen species with adjunctive therapies such as cognitive behavioral therapies have been proven to be effective in bipolar disorders.

Madireddy S, Madireddy S. Therapeutic interventions to mitigate mitochondrial dysfunction and oxidative stress-induced damage in patients with bipolar disorder. *Int J Mol Sci.* 2022;23(3):1844. doi:10.3390/ijms23031844

Laughter Yoga Reduces Perceptions of Stress

- 90 first-year nursing students in Turkey
- In the intervention group:
  - Eight 45-minute laughter yoga sessions implemented over four weeks.
  - Significant effects on perceived stress levels and improvement on meaning/purpose of life questionnaires compared to controls.

Kaya KÇ, Özkan ÇG, Ağiş D. Effects of laughter yoga practiced by the first year nursing students before clinical practice on their perceptions of stress and meaning of life: a randomized controlled trial. *BMC Nurs*. 2025;24(1):41. doi:10.1186/s12912-024-02642-0

Cognitive Behavior Therapy

Improved Glycemic Control in T2DM

Type 2 diabetic patients undergoing cognitive behavioral therapy (CBT) had significant improvements in glycemic control, diabetes self-care behavior, and fatigue compared to participants not utilizing CBT.

Alshehri MM, Alothman SA, Alenazi AM, et al. The effects of cognitive behavioral therapy for insomnia in people with type 2 diabetes mellitus, pilot RCT part II: diabetes health outcomes. *BMC Endocr Disord*. 2020;20(1):136. doi:10.1186/s12902-020-00612-6

PTSD and Pain in Active-Duty Service Members

- Patients with a PTSD diagnosis or positive screening experience:
  - Higher pain
  - Lower physical function
  - Reduced social satisfaction
  - Increased anger, anxiety, fatigue, and depression compared to those without PTSD ($p<0.0001$)

- Integrated PTSD treatment and pain neuroscience education are recommended for managing these comorbidities

Li H, Flynn DM, Highland KB, Barr PK, Langford DJ, Doorenbos AZ. Relationship between post-traumatic stress disorder symptoms and chronic pain-related symptom domains among military active duty service members. *Pain Med*. 2021;22(12):2876-2883. doi:10.1093/pm/pnab087

Weekly Religious Service Attendance

Decreased Odds of Dementia Diagnosis

- Individuals who never attended religious services had **2.46 times higher odds** of an Alzheimer's disease and related dementias diagnosis compared to those attending religious services more than once a week.
- Gender and education did not have significant effects on AD odds.

Data was concluded from the Health and Retirement Study, a representative longitudinal dataset with oversampling of Black adults.

Griffith EE, Robbins PA, Ferede BT, Bentley-Edwards KL. Religious participation is associated with fewer dementia diagnoses among Black people in the United States. *Am J Hum Biol*. 2024;36(12):e24125. doi:10.1002/ajhb.24125

Religious Involvement Has a Positive Association With Health Outcomes

- Heart Disease
- Hypertension
- Cerebrovascular Disease
- Dementia
- Immune Function
- Endocrine Function
- Cancer
- Overall Mortality

Reinert KG, Campbell JC, Bandeen-Roche K, Lee JW, Szanton S. The role of religious involvement in the relationship between early trauma and health outcomes among adult survivors. *J Child Adolesc Trauma*. 2016;9:231-241. doi:10.1007/s40653-015-0067-7

Religious Involvement and Health Outcomes

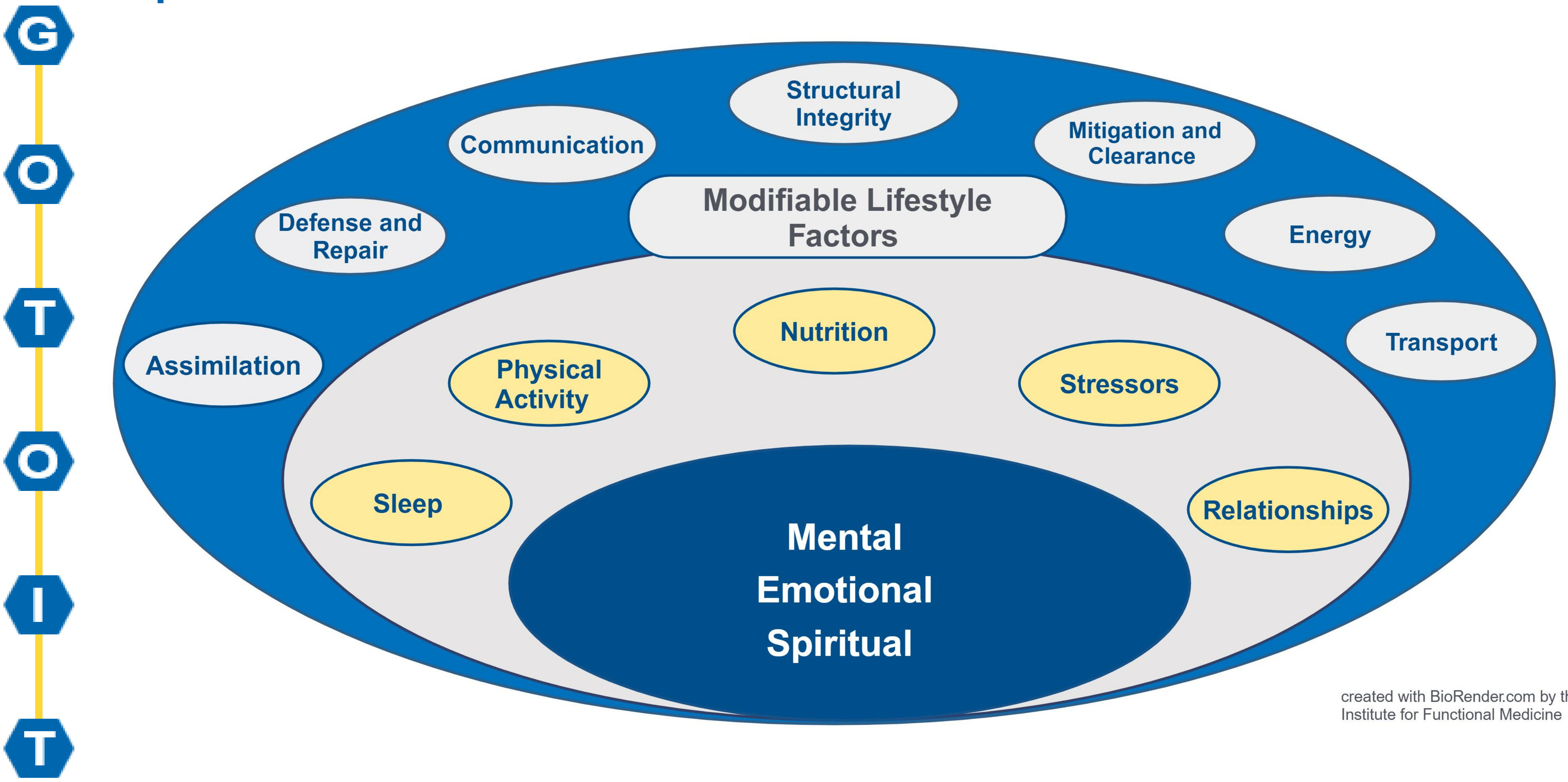
Early traumatic stress (ETS) has been shown to have a negative impact on mental and physical health.

Religious involvement indicators such as religious coping, intrinsic religiosity, gratitude, and forgiveness have a positive impact on health and may moderate the negative impact of ETS on mental health.

Furthermore, religious involvement may have a protective effect on mental health, especially the act of forgiveness.

Reinert KG, Campbell JC, Bandeen-Roche K, Lee JW, Szanton S. The Role of Religious Involvement in the Relationship Between Early Trauma and Health Outcomes Among Adult Survivors. *J Child Adolesc Trauma*. 2016;9:231-241. doi:10.1007/s40653-015-0067-7

Mental, Emotional, and Spiritual Factors
Epicenter of the Matrix and Patient Evaluation



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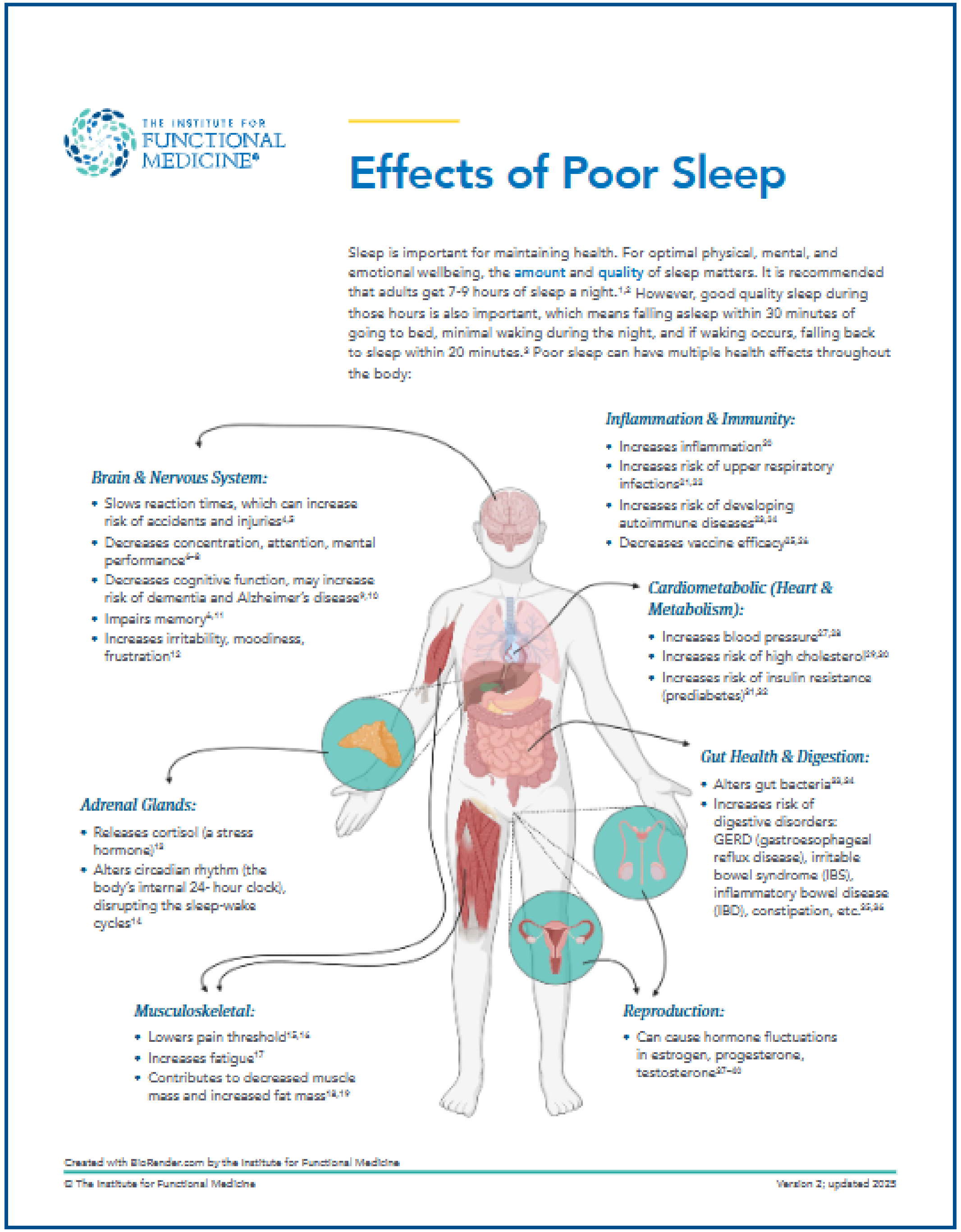
Long Description

- The GO TO IT framework and the functional medicine matrix elements with mental, emotional, and spiritual factors at the epicenter of the matrix.

Clinical Focus

- Patient-facing point-of-care tools to explain the effects of lifestyle behaviors on health.

MLFs and Physiological Function Sleep/Poor Sleep



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Effects of Poor Sleep

Sleep is important for maintaining health. For optimal physical, mental, and emotional wellbeing, the **amount** and **quality** of sleep matters. It is recommended that adults get 7-9 hours of sleep a night.[1,2] However, good quality sleep during those hours is also important, which means falling asleep within 30 minutes of going to bed, minimal waking during the night, and if waking occurs, falling back to sleep within 20 minutes.[3] Poor sleep can have multiple health effects throughout the body:

Brain & Nervous System:

- Slows reaction times, which can increase risk of accidents and injuries[4,5]
- Decreases concentration, attention, mental performance[6–8]
- Decreases cognitive function, may increase risk of dementia and Alzheimer's disease[9,10]
- Impairs memory[6,11]
- Increases irritability, moodiness, frustration[12]

Inflammation & Immunity:

- Increases inflammation[20]
- Increases risk of upper respiratory infections[21,22]
- Increases risk of developing autoimmune diseases[23,24]
- Decreases vaccine efficacy[25,26]

Cardiometabolic (Heart & Metabolism):

- Increases blood pressure[27,28]
- Increases risk of high cholesterol[29,30]
- Increases risk of insulin resistance (prediabetes)[31,32]

Adrenal Glands:

- Releases cortisol (a stress hormone)[13]
- Alters circadian rhythm (the body's internal 24- hour clock), disrupting the sleep-wake cycles[14]

Gut Health & Digestion:

- Alters gut bacteria[33,34]
- Increases risk of digestive disorders: GERD (gastroesophageal reflux disease), irritable bowel syndrome (IBS), inflammatory bowel disease (IBD), constipation, etc.[35,36]

Musculoskeletal:

- Lowers pain threshold[15,16]
- Increases fatigue[17]
- Contributes to decreased muscle mass and increased fat mass[18,19]

Reproduction:

- Can cause hormone fluctuations in estrogen, progesterone, testosterone[37–40]

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Long Description

- I F M Toolkit item: Effects of poor sleep.
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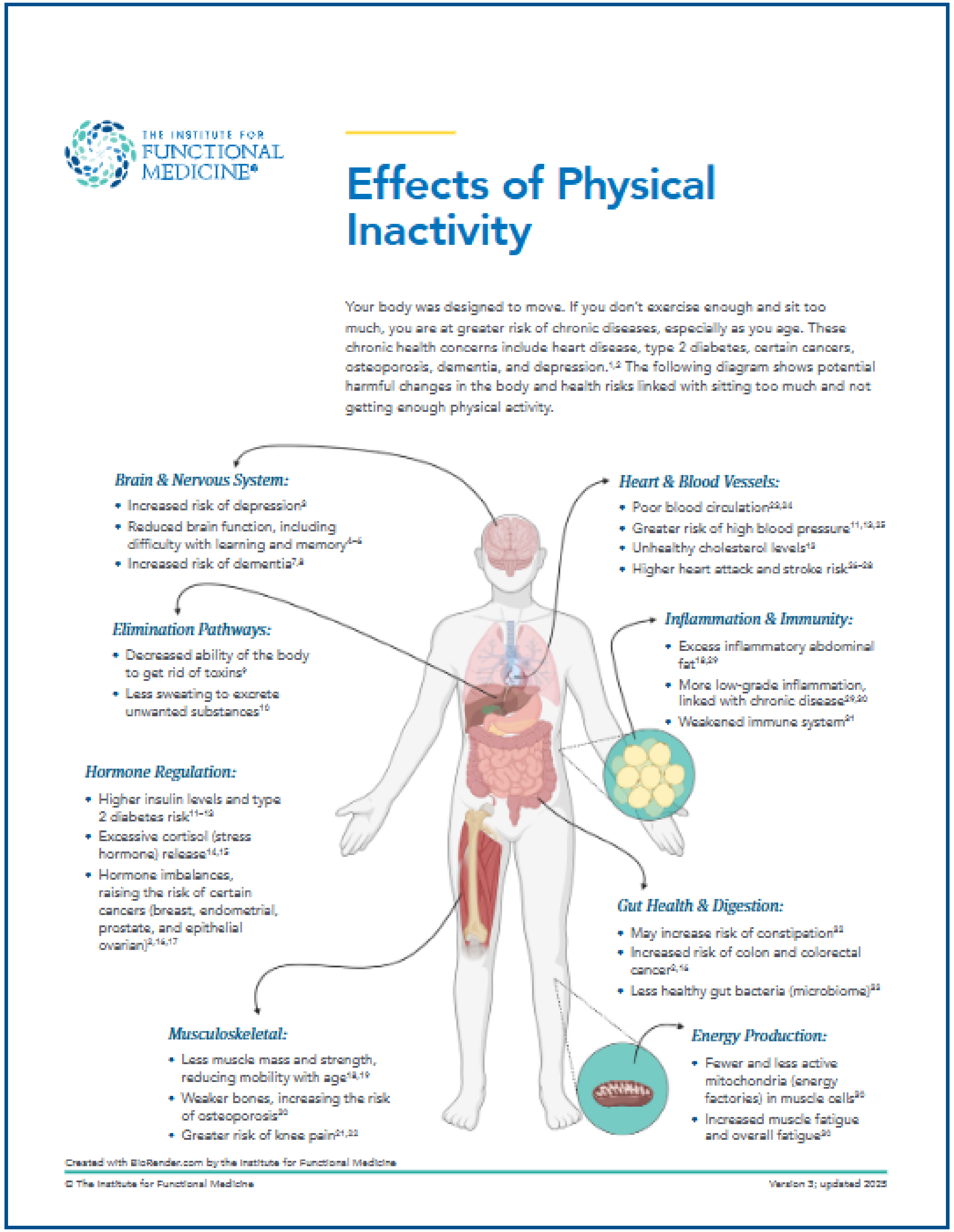
MLFs and Physiological Function Exercise/Sedentarism



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Effects of Physical Inactivity

Your body was designed to move. If you don't exercise enough and sit too much, you are at greater risk of chronic diseases, especially as you age. These chronic health concerns include heart disease, type 2 diabetes, certain cancers, osteoporosis, dementia, and depression.[1,2] The following diagram shows potential harmful changes in the body and health risks linked with sitting too much and not getting enough physical activity.

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Long Description

- I F M Toolkit item: Effects of physical inactivity.
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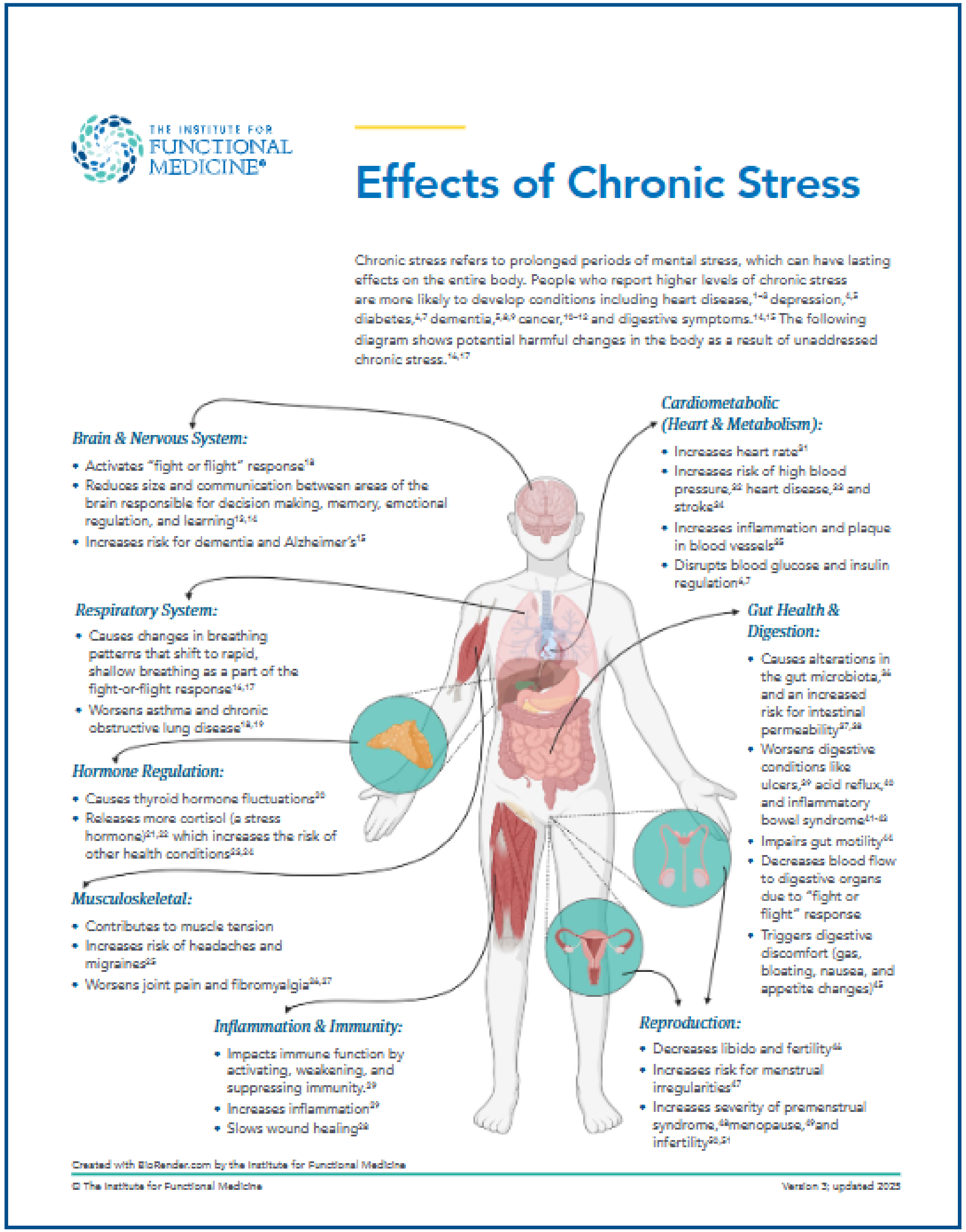
MLFs and Physiological Function Chronic Stress



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Effects of Chronic Stress

Chronic stress refers to prolonged periods of mental stress, which can have lasting effects on the entire body. People who report higher levels of chronic stress are more likely to develop conditions including heart disease,[1-3] depression,[4,5] diabetes,[6,7] dementia,[5,8,9] cancer,[10-13] and digestive symptoms.[14,15] The following diagram shows potential harmful changes in the body as a result of unaddressed chronic stress.[16,17]

Brain & Nervous System:

- Activates "fight or flight" response[18]
- Reduces size and communication between areas of the brain responsible for decision making, memory, emotional regulation, and learning[13,14]
- Increases risk for dementia and Alzheimer's[15]

Respiratory System:

- Causes changes in breathing patterns that shift to rapid, shallow breathing as a part of the fight-or-flight response[16,17]
- Worsens asthma and chronic obstructive lung disease[18,19]

Hormone Regulation:

- Causes thyroid hormone fluctuations[20]
- Releases more cortisol (a stress hormone)[21,22] which increases the risk of other health conditions[23,24]

Musculoskeletal:

- Contributes to muscle tension
- Increases risk of headaches and migraines[25]
- Worsens joint pain and fibromyalgia[26,27]

Inflammation & Immunity:

- Impacts immune function by activating, weakening, and suppressing immunity.[28]
- Increases inflammation[29]
- Slows wound healing[30]

Cardiometabolic (Heart & Metabolism):

- Increases heart rate[31]
- Increases risk of high blood pressure,[32] heart disease,[33] and stroke[34]
- Increases inflammation and plaque in blood vessels[35]
- Disrupts blood glucose and insulin regulation[6,7]

Gut Health & Digestion:

- Causes alterations in the gut microbiota,[36] and an increased risk for intestinal permeability[37,38]
- Worsens digestive conditions like ulcers,[39] acid reflux,[40] and inflammatory bowel syndrome[41-43]
- Impairs gut motility[44]
- Decreases blood flow to digestive organs due to "fight or flight" response
- Triggers digestive discomfort (gas, bloating, nausea, and appetite changes)[45]

Reproduction:

- Decreases libido and fertility[46]
- Increases risk for menstrual irregularities[47]
- Increases severity of premenstrual syndrome,[48] menopause,[49] and infertility[50,51]

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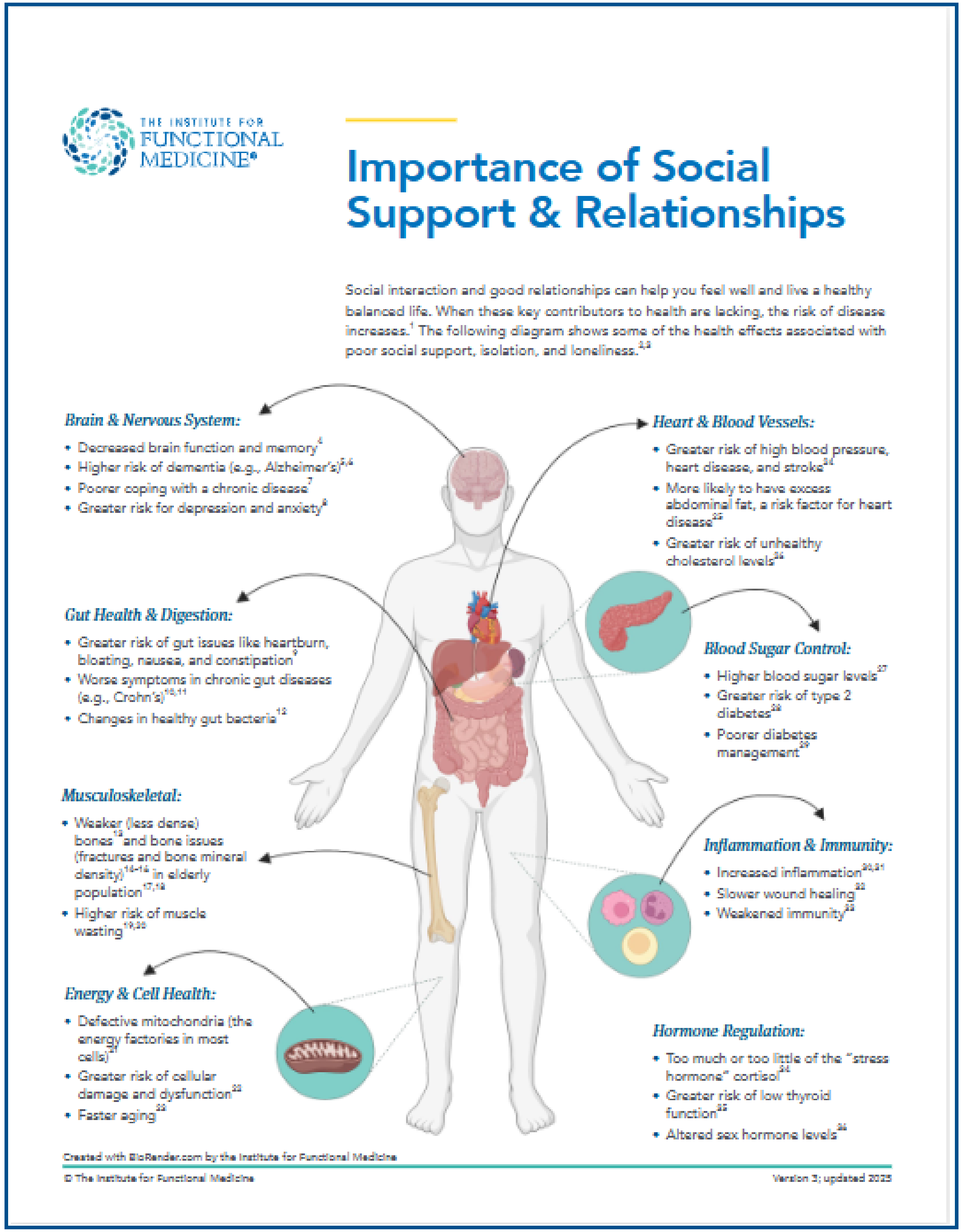
Long Description

- I F M Toolkit item: Effects of chronic stress
  - Accessible version available in course materials.

MLFs and Physiological Function Relationships



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Importance of Social Support & Relationships

Social interaction and good relationships can help you feel well and live a healthy balanced life. When these key contributors to health are lacking, the risk of disease increases.[1] The following diagram shows some of the health effects associated with poor social support, isolation, and loneliness.[2,3]

Brain & Nervous System:

- Decreased brain function and memory[4]
- Higher risk of dementia (e.g., Alzheimer's)[5,6]
- Poorer coping with a chronic disease[7]
- Greater risk for depression and anxiety[8]

Gut Health & Digestion:

- Greater risk of gut issues like heartburn, bloating, nausea, and constipation[9]
- Worse symptoms in chronic gut diseases (e.g., Crohn's)[10,11]
- Changes in healthy gut bacteria[12]

Musculoskeletal:

- Weaker (less dense) bones[13] and bone issues (fractures and bone mineral density)[14-16] in elderly population[17,18]
- Higher risk of muscle wasting[19,20]

Energy & Cell Health:

- Defective mitochondria (the energy factories in most cells)[21]
- Greater risk of cellular damage and dysfunction[22]
- Faster aging[23]

Heart & Blood Vessels:

- Greater risk of high blood pressure, heart disease, and stroke[24]
- More likely to have excess abdominal fat, a risk factor for heart disease[25]
- Greater risk of unhealthy cholesterol levels[26]

Blood Sugar Control:

- Higher blood sugar levels[27]
- Greater risk of type 2 diabetes[28]
- Poorer diabetes management[29]

Inflammation & Immunity:

- Increased inflammation[30,31]
- Slower wound healing[32]
- Weakened immunity[33]

Hormone Regulation:

- Too much or too little of the "stress hormone" cortisol[34]
- Greater risk of low thyroid function[35]
- Altered sex hormone levels[36]

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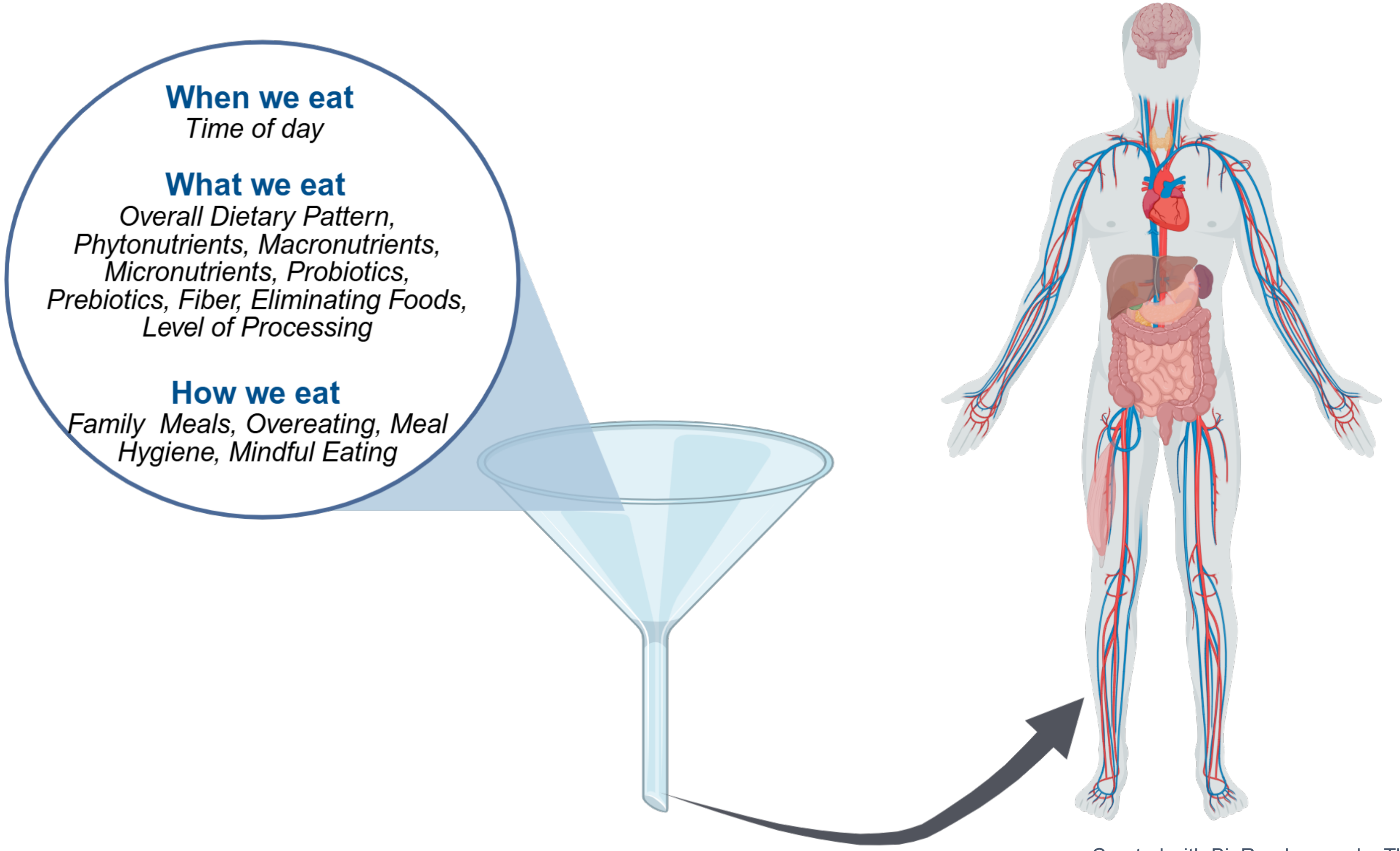
Long Description

- I F M Toolkit item: Importance of social support and relationships
  - Accessible version available in course materials.

MLFs and Physiological Function Nutrition



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Long Description

- When we eat, what we eat, and how we eat all have an impact on physiological function.

Clinical Focus

- Summary, conclusion, and key clinical takeaways.

Modifiable Lifestyle Factors: Clinical Takeaways

Lifestyle behaviors modulate physiological function across bodily systems and corresponding nodes of the matrix.

Modifiable lifestyle factors are often antecedents and/or mediators of dysfunction on each node of the matrix.

IFM has provided you with access to patient educational tools that will help your patients successfully initiate lifestyle behavior changes.

Modifiable Lifestyle Factors: Clinical Takeaways

Modifiable lifestyle factors are often ideal initial therapeutic interventions because they directly address current mediators of dysfunction.

Modifiable lifestyle factors have excellent profiles with regard to safety and cost-effectiveness.

To optimize motivation, feasibility, and outcomes, lifestyle behavior changes need to be co-prioritized with the patient and informed by their goals, preferences, context, & circumstances.

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Modifiable Lifestyle Factors

Mechanisms of Action on Physiological Function

APPLYING FUNCTIONAL MEDICINE IN CLINICAL PRACTICE